A Grid of Numbers · AS · Practice

U09b · 24 written + 6 pen-first = 30 marks · Name: _____ Due: _____

Fresh authored questions. Show working for calculations. Unless stated otherwise, calculate packed, uncompressed pixel data only: no header, palette, padding or compression. 1 byte = 8 bits; 1 KiB = 1024 bytes; 1 MiB = 1024 KiB.

For decimal units: 1 MB = 1 000 000 bytes.

1. Ours: samples and grids

(a) State what is meant by colour depth in a bitmap image. **[1]**

(b) State the maximum number of colours representable with 6 bits per pixel. **[1]**

(c) Distinguish image resolution from screen resolution. **[2]**

2. Ours: read the unit

(a) An image is 900 × 600 pixels at **4 bytes per pixel**. Calculate its pixel-data size in MB. Show working. **[2]**

(b) Another image is 768 × 512 pixels at 16 bits per pixel. Calculate its pixel-data size in MiB. Show working. **[2]**

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3. Ours: interpret the file

(a) A bitmap header already records width and height. State two other items it may store to help a reader interpret the image. **[2]**

(b) A file format in this question stores a 128-byte header followed by exactly 4096 bytes of uncompressed pixel data, with no other data. Calculate the total file size. State why a compressed version need not have the same size. **[2]**

4. Ours: a drawing recipe

A vector drawing contains a blue rectangle and a red circle. The circle is drawn after the rectangle and overlaps it.

(a) Identify one drawing object in this description. State one property, other than colour, that can be stored for it. **[2]**

(b) Describe a drawing list and explain how swapping the two objects in this list changes the overlap. **[2]**

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5. Ours: justify the representation

(a) A wildlife website needs a detailed photograph of a bird. Choose bitmap or vector as a suitable primary representation and justify your choice from the image content. **[2]**

(b) A club badge consists of a few circles, straight lines and text. It will be printed on both a pen and a large banner. Choose a representation and justify it from this task. **[2]**

6. Ours: changes multiply

(a) For a new capture, width is doubled, height is tripled and colour depth rises from 8 to 16 bits per pixel. Calculate the pixel-data size multiplier. Show working. **[2]**

(b) Explain one potential quality benefit of the higher colour depth at capture, and why converting an old 8-bit image to 16 bits per pixel need not provide that benefit. **[2]**

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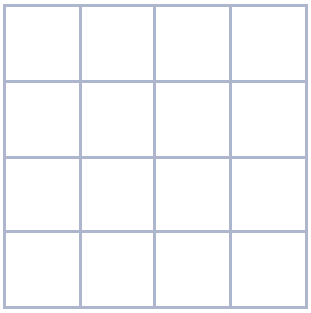
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7. Ours: reconstruct a tiny picture · pen first [6]

💻 **Design, then trace.** No code is required. A 4 × 4 one-bit bitmap stores rows from top to bottom, each row left to right. 1 means black; 0 means white. Its packed pixel bits are **1000 0100 0010 0001**. These six marks are human marked.

(a) Write numbered steps to turn the supplied bits into a picture. Include when to start a new row. **[2]**

(b) Trace your steps: shade the grid to reconstruct the picture. **[2]**



(c) A separate RGB pixel has three 8-bit channels: R = 12, G = 128, B = 225. Write its six-digit hexadecimal colour as #RRGGBB. Preserve leading zeroes. **[2]**

Above and beyond · outside the 30 marks. Two 2 × 2 black/white patches have rows **01 / 10** and **00 / 11**. A resizer replaces each patch with its mean value (black = 1, white = 0). Predict the two outputs. Can that single output uniquely identify which original patch was used? Explain.

Additional space - label the question number.